DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-100-05 — Replace VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/A2)

Area	A100 Tank farm	**Task**	VR Valve replacement in kind
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	**T-102** — T-102 tank 12' dia × 22' high — E 44'-0" N 26'-6" EL 100'-6"		
Equipment	T-102	**Lines**	2"-P150-SC-3010
Isometrics	ISO-2"-P150-SC-3010 (generated); owner's drawings ISO-SC-3010		
Permits	General work permit		

Reason

Condensate isolation valve passing; replace in kind.

Scope of work

1. Isolate per the isolation list, drain and vent 2"-P150-SC-3010 between the isolations.
2. Unbolt and remove VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/A2) at E 37'-5/8" N 26'-6" EL 104'-6".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, GATE, WEDGE, OS&Y 2" P150 of the same size, class and face-to-face (GATE2-P150), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 2"-P150-SC-3010/A2, 2"-P150-SC-3010/A1, 2"-P150-SC-3010/A4

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-SC-3010	2"-P150-SC-3010	2"-P150-SC-3010	1	0 error(s), 1 warning(s) — see below

ISO-SC-3010 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-SC-3010	1	warning	FROM label 'T-102 NOZ. N8' is not on the drawing

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-102	equipment	T-102 tank 12' dia × 22' high		E 44'-0" N 26'-6" EL 100'-6"
2"-P150-SC-3010/A2	item	VALVE, GATE, WEDGE, OS&Y 2" P150	2"-P150-SC-3010	E 37'-5/8" N 26'-6" EL 104'-6"
2"-P150-SC-3010/A1	item	GASKET 2" P150 COMPRESSED SHEET	2"-P150-SC-3010	E 37'-6-1/16" N 26'-6" EL 104'-6"
2"-P150-SC-3010/A4	item	GASKET 2" P150 COMPRESSED SHEET	2"-P150-SC-3010	E 36'-7-3/16" N 26'-6" EL 104'-6"
T-102-N8	nozzle	Nozzle T-102-N8 (condensate out, 2" P150) on T-102		E 37'-6-1/8" N 26'-6" EL 104'-6"
2"-P150-SC-3010/B3	item	VALVE, GATE, WEDGE, OS&Y 2" P150	2"-P150-SC-3010	E 34'-0" N -2'-0" EL 123'-6-3/16"
2"-P150-SC-3010/B6	item	FLANGE WELD-NECK 2" P150 RAISED FACE	2"-P150-SC-3010	E 34'-0" N -2'-0" EL 124'-1-13/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	T-102-N8	2"-P150-SC-3010	Line end at nozzle T-102-N8: the equipment is out of service, drained and isolated under the shutdown isolation			
2	valve	2"-P150-SC-3010/B3	2"-P150-SC-3010	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/B3)	E 34'-0" N -2'-0" EL 123'-6-3/16"		
3	blind	2"-P150-SC-3010/B6	2"-P150-SC-3010	Fit a line blind between 2"-P150-SC-3010/B3 and FLANGE WELD-NECK 2" P150 RAISED FACE (2"-P150-SC-3010/B6), work side	E 34'-0" N -2'-0" EL 123'-11-11/16"		

Parts

Description	Qty	Replaces
VALVE, GATE, WEDGE, OS&Y 2" P150	1	2"-P150-SC-3010/A2
GASKET 2" P150 COMPRESSED SHEET	1	2"-P150-SC-3010/A1
STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150	1	2"-P150-SC-3010/A3
GASKET 2" P150 COMPRESSED SHEET	1	2"-P150-SC-3010/A4
STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150	1	2"-P150-SC-3010/A6

Inspections

Id	Inspection	Criteria	Basis
WO-100-05-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
2"-P150-SC-3010/A1	4 × 0.625" studs (2"-P150-SC-3010/A3)	13,806	3	
2"-P150-SC-3010/A4	4 × 0.625" studs (2"-P150-SC-3010/A6)	13,806	3	

Test — Service leak test

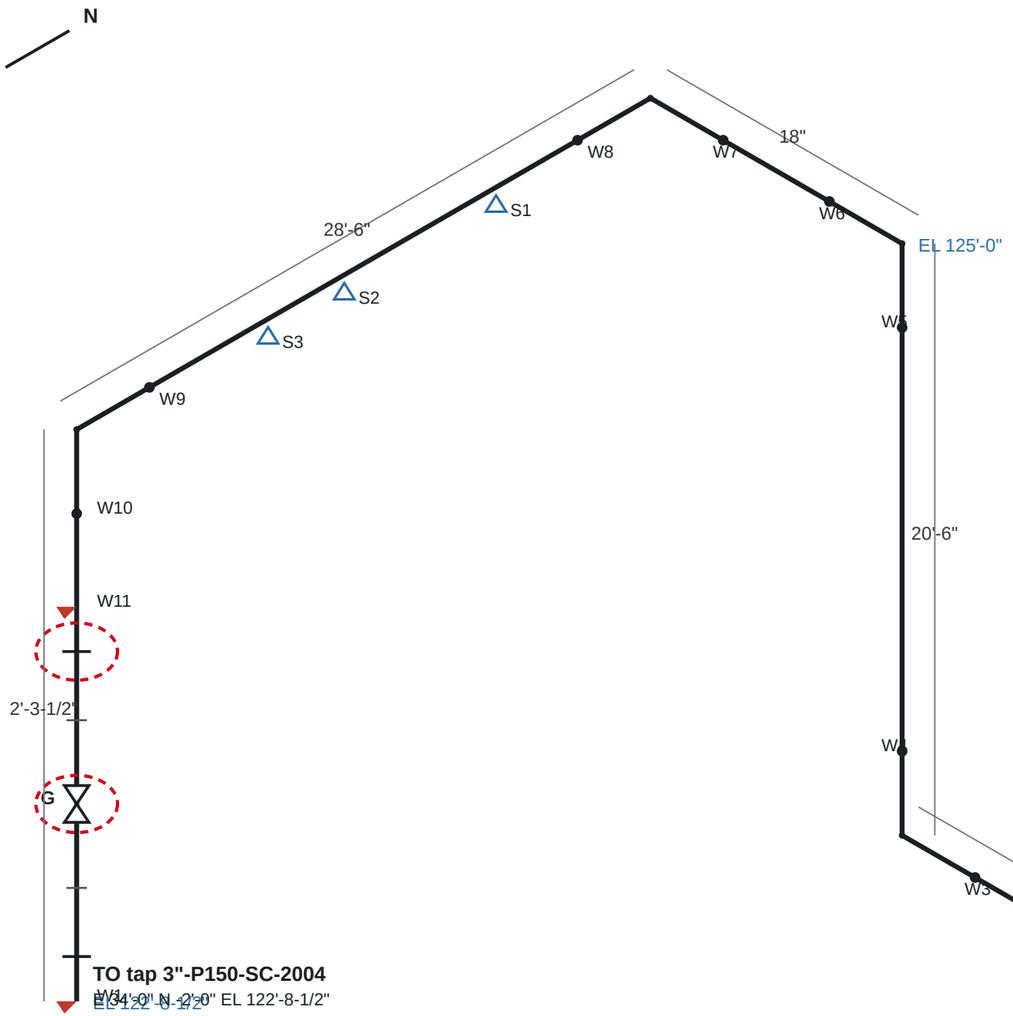
Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 2"-P150-SC-3010/A2 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	GATE2-P150	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 2"-P150-SC-3010 — Replace VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/A2)



BILL OF MATERIALS

1	GASKET 2" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 2" P150	2
3	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150	4
4	FLANGE WELD-NECK 2" P150 RAISED FACE	3
5	PIPE 2" P150 BEVELLED ENDS	49'-8-3/8"
6	ELBOW 90° R = 5/3 OD 2" P150	4

WELDS 11 (3 FIELD) · SUPPORTS 3 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

2"-P150-SC-3010

CLASS C150 — Steam condensate — INSULATED
SERVICE SC · 2" · P150 · DESIGN 150 psi
T-102-N8 → tap 3"-P150-SC-2004
PIPE 49'-8-3/8" · WEIGHT 284 lb

DWG ISO-2"-P150-SC-3010 REV 0 WO-100-05

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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